

grep Cheat Sheet

Search text for lines that match a pattern.

Using it

```
grep [options] 'pattern' file... ← search files
command | grep 'pattern' ← search another command's output
```

Option	What it does	Option	What it does
-i	Ignores upper/lower case	-r	Searches folders, including subfolders
-v	Shows lines that do not match	-l	Lists only the names of matching files
-n	Shows line numbers	-L	Lists files with no match
-c	Counts matching lines	-h	Hides file names in the output
-w	Matches whole words only	-o	Shows only the matching part
-x	Matches whole lines only	-q	Prints nothing; only sets the exit status
-E	Extended patterns (+ ? () {})	-A 3	Also shows 3 lines after each match
-F	Treats the pattern as plain text	-B 3	Also shows 3 lines before each match
-e pat	Adds a pattern (repeatable)	-C 3	Also shows 3 lines before and after
--include	--include= '*.conf' limits -r	--exclude-dir	--exclude-dir= .git skips a folder

Pattern basics

Pattern	Matches	Pattern	Matches
.	Any single character	[abc]	One of a, b or c
^	Start of the line	[^abc]	Any character except a, b or c
\$	End of the line	[0-9]	Any digit ([a-z] = any lowercase letter)
*	The previous item, 0 or more times	a b *	a or b
+ *	The previous item, 1 or more times	(ab)+ *	Groups items together
? *	The previous item is optional	{2,4} *	The previous item, 2 to 4 times
\.	A literal dot (\ cancels special meaning)	\b	A word boundary

* needs -E

Reading the output

```
$ grep -rn -C1 'error' config/
config/app.conf-11-[logging] ← "-" separators = context line (from -A/-B/-C)
config/app.conf:12:level = error ← ":" separators = the matching line (file:line:text)
config/app.conf-13-file = /var/log/app.log
-- ← separates groups of matches that aren't next to each other
config/db.conf:4:on_error = retry
```

The file name only appears when searching more than one file (or with **-r**). Exit status: **0** = found, **1** = not found, **2** = error.

Handy recipes

Command	Does
grep -rn 'TODO' .	Finds TODO in every file under the current folder, with line numbers
grep -Ei 'error warn' app.log	Finds lines containing error or warn, any case
grep -v '^#' file grep -v '^\$'	Hides comment lines and blank lines
grep -c 'Failed' /var/log/auth.log	Counts failed logins
ps aux grep '[n]ginx'	Finds a process without matching the grep itself
grep -oE '[0-9]+(\.[0-9]+){3}' log	Pulls out just the IPv4 addresses
grep -rl 'oldname' src/	Lists files that still mention oldname

Remember: put patterns in **single quotes** so the shell leaves them alone. Searching for text with **.** ***** **[** **\$** in it? Use **-F** to match it literally.